Wednesday Warm Up: Unit 1, Capacitors, current and DC circuits

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1 Memory Bank

- $\vec{E}_{\rm plane} = \frac{\sigma}{2\epsilon_0}\hat{k} = \frac{Q}{2A\epsilon_0}\hat{k}$ ... The electric field of a plate or disk with charge Q and area A, for distances much smaller than the radius or length.
- $\Delta U = q\Delta V$ ... Relationship between change in potential energy, charge, and voltage.
- $V(r) = kq/r$ ... Potential (voltage) of a point charge.
- $V(x) = Ex + V_0$ ... Linear relationship between voltage, electric field, and distance between two charge plates to form a capacitor.
- $Q = C\Delta V$ ... Relationship between stored charge, capacitance, and voltage.
- $I = \Delta Q/\Delta t$ ... The relationship between current, I, and charge.
- $C_{\rm tot}^{-1} = C_1^{-1} + C_2^{-1}$ ... Two capacitors *in series*.
- $C_{\rm tot} = C_1 + C_2$ ... Two capacitors *in parallel*.
- $V_{\rm C}(t) = \mathcal{E}\left(1-\exp(-t/\tau)\right)$ ... The capacitor **charging** voltage in an RC circuit, with $\tau = RC$.
- $V_{\rm C}(t) = V_{\rm C,i}\exp(-t/\tau)$ ... The capacitor **discharging** voltage in an RC circuit, with $\tau = RC$.
- $\Delta U = Q^2/(2C) = \frac{1}{2}CV^2$ ... Energy stored in a capacitor.

2 Capacitors and Current

1. (a) Two 1 μF capacitors are connected *in parallel*. What is the total capacitance? (b) If we charge the capacitors with 12 V, how much charge is stored on the capacitors? (c) How much energy is stored in the circuit?
2. Repeat the above exercise, but assume the capacitors are *in series*.
3. Suppose we connect a headlight to a car battery with charge 80 Amp hr. (a) If the headlight draws 0.5 Amp of current, how long will the battery last? (b) How much charge flowed through the headlight?
4. The timing device in an automobile's intermittent wiper system is based on an RC time constant and utilizes a 0.500 microfarad capacitor and a variable resistor. Over what range must R be made to vary to achieve time constants from 2.00 to 15.0 s?
5. A heart defibrillator being used on a patient has an RC time constant of 10.0 ms due to the resistance of the patient and the capacitance of the defibrillator. (a) If the defibrillator has an 8.00 microfarad capacitance, what is the resistance R of the path through the patient? (b) If the initial voltage is 12.0 kV, how long does it take to decline to 600 V?
6. In the previous exercise, how much energy is delivered to the patient's heart?